

Problem 25. Let $n \geq 2$, and let $a_1, a_2, \dots, a_n$ be real numbers such that $a_j > 1$ for $j = 1, 2, \dots, n$, and $|a_j - a_{j+1}| < 1$ for $j = 1, 2, \dots, n-1$. Prove that

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} < 2n - 1.$$

Solution. We prove the inequality by induction on n .

When $n = 2$, we need to prove $\frac{a_1}{a_2} + \frac{a_2}{a_1} < 3$, which is equivalent to $\frac{(a_1 - a_2)^2}{a_1 a_2} < 1$. This holds since $(a_1 - a_2)^2 < 1$ and $a_1 a_2 > 1$.

Assume the case $n - 1$ holds, i.e.

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-2}}{a_{n-1}} + \frac{a_{n-1}}{a_1} < 2n - 3.$$

Consider the case n . Then we have

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} = \left(\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-2}}{a_{n-1}} + \frac{a_{n-1}}{a_1} \right) + \frac{a_{n-1}}{a_n} + \frac{a_n - a_{n-1}}{a_1} < 2n - 3 + \frac{a_{n-1}}{a_n} + \frac{a_n - a_{n-1}}{a_1}$$

by the inductive hypothesis.

- If $a_{n-1} \leq a_n$, then $\frac{a_{n-1}}{a_n} \leq 1$. Also, we have $\frac{a_n - a_{n-1}}{a_1} < \frac{1}{1} = 1$. So we obtain

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} < 2n - 3 + 1 + 1 = 2n - 1.$$

- If $a_{n-1} > a_n$, then $\frac{a_n - a_{n-1}}{a_1} < 0$. Also, we have $\frac{a_{n-1}}{a_n} < \frac{a_n + 1}{a_n} < 2$ since $a_n > 1$. It follows that

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} < 2n - 3 + 2 = 2n - 1.$$

Thus, we have proven the inequality in any case. So we are done by induction. □

Problem 26 (Canada MO 2014 Problem 3). Let p be an odd prime. How many tuples $(x_1, x_2, \dots, x_p)$ are there with $x_j \in \{0, 1, \dots, p-1\}$ for each j , such that $p \mid x_1 x_2 + x_2 x_3 + x_3 x_4 + \dots + x_{p-1} x_p + x_p x_1$ and $p \nmid x_1 + x_2 + \dots + x_p$?

Solution. The answer is $(p-1)p^{p-2}$.

There are $(p-1)p^{p-1}$ tuples $(x_1, x_2, \dots, x_p)$ such that $x_j \in \{0, 1, \dots, p-1\}$ for each j , and $p \nmid x_1 + x_2 + \dots + x_p$. This is because we can choose $x_1, x_2, \dots, x_{p-1}$ in any ways (p choices for each term), and choose x_p in $p-1$ ways (since there is a unique x_p satisfying $p \mid x_1 + x_2 + \dots + x_p$).

These tuples can be partitioned into $(p-1)p^{p-2}$ groups such that each group consists of p tuples of the form

$$(x_1, x_2, \dots, x_p), (x_1 + 1, x_2 + 1, \dots, x_p + 1), \dots, (x_1 + p - 1, x_2 + p - 1, \dots, x_p + p - 1).$$

(Note that if $p \nmid x_1 + x_2 + \dots + x_p$, then $p \nmid (x_1 + k) + (x_2 + k) + \dots + (x_p + k)$ for any k .) Now, note that

$$\begin{aligned} & (x_1 + k)(x_2 + k) + (x_2 + k)(x_3 + k) + \dots + (x_{p-1} + k)(x_p + k) + (x_p + k)(x_1 + k) \\ &= (x_1 x_2 + x_2 x_3 + x_3 x_4 + \dots + x_{p-1} x_p + x_p x_1) + 2(x_1 + x_2 + \dots + x_p)k + pk^2 \\ &\equiv (x_1 x_2 + x_2 x_3 + x_3 x_4 + \dots + x_{p-1} x_p + x_p x_1) + 2(x_1 + x_2 + \dots + x_p)k \pmod{p}. \end{aligned}$$

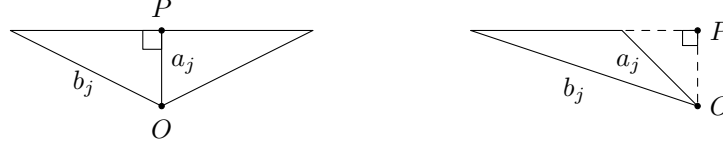
Since $p \nmid x_1 + x_2 + \dots + x_p$ and p is odd, there exists a unique k such that this is divisible by p . In other words, exactly one tuple in each group satisfies $p \mid x_1 x_2 + x_2 x_3 + x_3 x_4 + \dots + x_{p-1} x_p + x_p x_1$. This proves there are $(p-1)p^{p-2}$ desired tuples. □

Problem 27 (Belarus MO 2012 Problem 4). There are $n \geq 2$ segments inside a unit circle with centre O such that the total length of the segments is $2\sqrt{n}$. Prove that there exists a circle with centre O that meets at least two of the segments (possibly meeting at the endpoint).

Solution. For the j th segment, let a_j and b_j be the shortest and the longest distance from O to this segment respectively. Then a circle with centre O and radius r meets this segment if and only if $a_j \leq r \leq b_j$. Thus, suppose on the contrary that $0 \leq a_1 < b_1 < a_2 < b_2 < \dots < a_n < b_n \leq 1$.

Let P be the projection of O onto the j th segment. If P lies on this segment, then we have $a_j = OP$, and the length of this segment is at most $2\sqrt{b_j^2 - a_j^2}$. If P lies outside this segment, then $OP < a_j$, and so the length of this segment is

$$\sqrt{b_j^2 - OP^2} - \sqrt{a_j^2 - OP^2} = \frac{b_j^2 - a_j^2}{\sqrt{b_j^2 - OP^2} + \sqrt{a_j^2 - OP^2}} \leq \frac{b_j^2 - a_j^2}{\sqrt{b_j^2 - a_j^2}} < 2\sqrt{b_j^2 - a_j^2}.$$



Therefore, the total length of the segments is at most

$$2 \sum_{j=1}^n \sqrt{b_j^2 - a_j^2} \leq 2 \sqrt{n \sum_{j=1}^n (b_j^2 - a_j^2)} = 2 \sqrt{n \left(b_n^2 - \sum_{j=1}^{n-1} (a_{j+1}^2 - b_j^2) - a_1^2 \right)} < 2 \sqrt{n(1^2 - 0^2)} = 2\sqrt{n}.$$

This is a contradiction. So there must be a circle with centre O that meets at least two of the segments □

Problem 28 (239 Open MO 2016 Senior Problem 6). Let $S = \{1, 2, \dots, n\}$, and let $\mathcal{F}$ be a family of subsets of S . It is known that for any $A, B \in \mathcal{F}$, we have $A \cup B \in \mathcal{F}$. Also, for any $A \in \mathcal{F}$, we have $3 \nmid |A|$. Prove that there exist $x, y \in S$ such that every A in $\mathcal{F}$ contains at least one of x and y .

Solution. Note that increasing the number of elements in S would not change the assertion. WLOG assume $n \equiv 2 \pmod{3}$. For any subset A of S , let A' be the set of all unordered pairs $\{x, y\}$ satisfying $x, y \in S$, $x \neq y$ and $A \cap \{x, y\}$ is nonempty. Note that the assertion is to prove that $A'_1 \cap A'_2 \cap \dots \cap A'_k$ is nonempty where $A_1, A_2, \dots, A_k$ are all the sets in $\mathcal{F}$. Indeed, we shall prove by induction that $|A'_1 \cap A'_2 \cap \dots \cap A'_m| \equiv 1 \pmod{3}$ for any m , which definitely implies the intersection is nonempty.

When $m = 1$, let $|A_1| = a \equiv 1, 2 \pmod{3}$. By definition, we easily see that

$$|A'_1| = \binom{a}{2} + a(n - a) = \frac{a[(a - 1) + (2n - 2a)]}{2} \equiv 2a(2n - 1 - a) \equiv -2a^2 \equiv 1 \pmod{3}.$$

Next, assume $|A'_1 \cap A'_2 \cap \dots \cap A'_j| \equiv 1 \pmod{3}$ for every $j < m$. Consider the case $j = m$. By the inclusion-exclusion principle, we have

$$|A'_1 \cup A'_2 \cup \dots \cup A'_m| = \sum_{i=1}^m |A'_i| - \sum_{1 \leq i < j \leq m} |A'_i \cap A'_j| + \dots + (-1)^{m+1} |A'_1 \cap A'_2 \cap \dots \cap A'_m|.$$

By the inductive hypothesis, each term on the right except possibly the last term is congruent to 1 modulo 3. For the left-hand side, it is easy to see by definition that $A'_1 \cup A'_2 \cup \dots \cup A'_m = (A_1 \cup A_2 \cup \dots \cup A_m)'$ (because $A_i \cap \{x, y\}$ is nonempty for some i if and only if $(A_1 \cup A_2 \cup \dots \cup A_m) \cap \{x, y\}$ is nonempty). By the given condition, $A_1 \cup A_2 \cup \dots \cup A_m$ is also a set in $\mathcal{F}$, and so it has size congruent to 1 modulo 3. It follows that

$$(-1)^{m+1} |A'_1 \cap A'_2 \cap \dots \cap A'_m| \equiv 1 - \binom{m}{1} + \binom{m}{2} - \dots + (-1)^m \binom{m}{m-1} = (1 - 1)^m - (-1)^m = (-1)^{m+1} \pmod{3}.$$

This proves $|A'_1 \cap A'_2 \cap \dots \cap A'_m| \equiv 1 \pmod{3}$. The result follows by induction. □